Session C6: Chemical Reactions I — Evidence, Equations, Balancing and the Five Reaction Types

Duration: 45 minutes
Continues from: Sessions C1–C5. The teacher's Units 1–3 (atomic structure, periodic trends, chemical bonding) are done. Writing correct formulas from names (C1, C2) is the skill this unit is built on: an equation with a wrong formula can't be balanced correctly.
School unit: "Chemical Reactions and Equations", the first unit after bonding. Stoichiometry (moles and mass relationships) follows it.
Grade-9/10 foundation: NCERT Class 9 *Atoms and Molecules* (the law of conservation of mass); NCERT Class 10 Ch1 *Chemical Reactions and Equations* (writing and balancing equations; combination, decomposition, displacement and double displacement reactions).
Grade-11 depth: NCERT Class 11 Ch1 *Some Basic Concepts of Chemistry* (laws of chemical combination). US textbook equivalent: the "Chemical Equations and Reactions" chapter of HMH *Modern Chemistry* (describing reactions, the types of reactions, the activity series).

NCERT and US names for the same reactions:

NCERT (India)	US (Dhairya's school)
combination	synthesis
decomposition	decomposition
displacement	single replacement
double displacement	double replacement
(burning, oxidation)	combustion

Lesson Plan

Objectives. By the end of this session Dhairya should be able to:

1. Tell a chemical change from a physical one, and list the five kinds of evidence for a chemical reaction.
2. State the **law of conservation of mass** and explain it with atoms.
3. Turn a word equation into a skeleton equation, then **balance** it by changing coefficients only, checking with an atom-count table.
4. Classify a reaction as **synthesis, decomposition, single replacement, double replacement or combustion**.
5. **Predict the products** of each type, using the activity series and the solubility rules.

Time	Segment
0–4 min	Warm-up: write 5 formulas from names
4–10 min	Part 1: physical vs chemical change, evidence of a reaction

Time	Segment
10–25 min	Part 2: conservation of mass, writing and balancing equations
25–40 min	Part 3: the five reaction types and predicting products
40–45 min	Exit ticket, assign homework

Pacing note. Part 2 is the core skill: do all four worked examples even if Part 3 gets shorter. If time is short, do synthesis, decomposition and combustion in class and set the activity series and solubility rules (single and double replacement) as reading with homework Q13 and Q14; they come back at the start of C7.

Warm-up (4 min)

Write the formula of each compound.

1. calcium chloride → **$CaCl_2$** (Ca^{2+} and Cl^-: two Cl^- balance one Ca^{2+})
2. aluminum sulfate → **$Al_2(SO_4)_3$** (Al^{3+} and SO_4^{2-}: $2 \times (+3) + 3 \times (-2) = 0$, and the parentheses keep the polyatomic ion together)
3. dinitrogen tetroxide → **N_2O_4** (molecular: the prefixes give the numbers)
4. sodium hydroxide → **$NaOH$**
5. iron(III) oxide → **Fe_2O_3** (the Roman numeral gives Fe^{3+})

Bridge to today: every equation below starts from formulas like these. Once a formula is correct, its subscripts are fixed. Balancing never changes them.

Part 1 — Physical vs Chemical Change (6 min)

A **physical change** alters the form or state of a substance but not its identity: melting ice is still H_2O. A **chemical change** (a chemical reaction) produces **new substances** with new properties: iron rusting becomes iron oxide.

Evidence that a chemical reaction has happened

Evidence	Example
A **gas** is produced (bubbles) that wasn't there before	zinc in hydrochloric acid gives hydrogen gas
A **precipitate** forms: a solid appears when two solutions are mixed	lead(II) nitrate + potassium iodide gives yellow solid PbI_2
A **color change**	a copper roof turning green
An **energy change**: the mixture gets hot or cold, or gives off light	a burning candle; an instant cold pack
A new **odor**	milk going sour

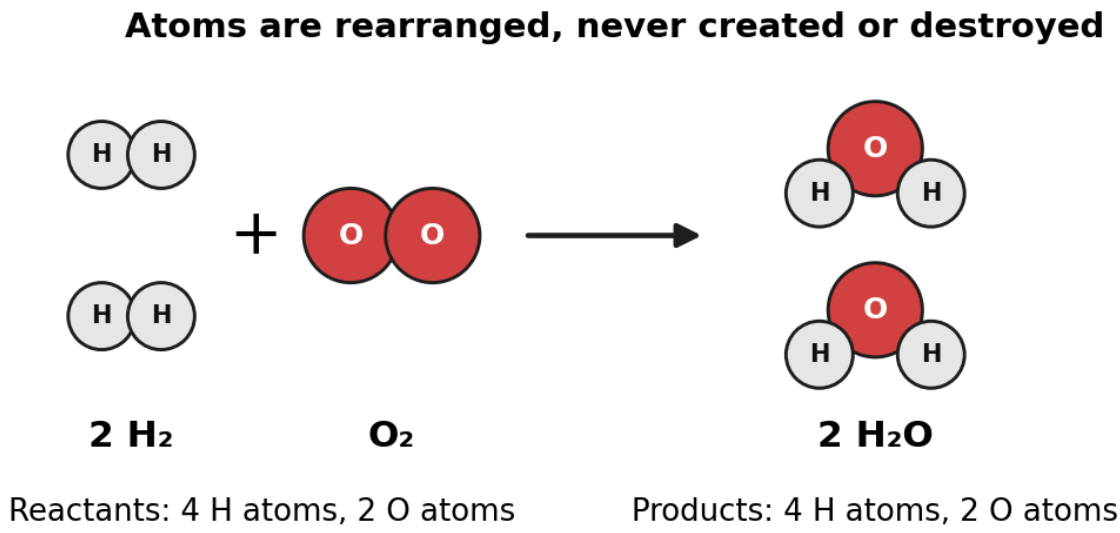
Careful: evidence is a clue, not proof. Boiling water makes bubbles, but the gas is still H_2O, so boiling is physical. Dissolving salt in water can feel slightly cold, yet no new substance forms. The real test is always: **is a new substance made?**

Part 2 — Conservation of Mass, Writing and Balancing Equations (15 min)

The law of conservation of mass (Lavoisier, 1789)

> **Law of conservation of mass.** In a chemical reaction, matter is neither created nor destroyed: the total mass of the products equals the total mass of the reactants.
>
> $$\text{mass of reactants} = \text{mass of products}$$

Why it is true. A reaction breaks bonds and makes new ones. The **atoms** are rearranged into new substances, but every atom that goes in comes out. Same atoms, same mass.



Example. Heating 10.0 g of calcium carbonate gives 5.60 g of calcium oxide and carbon dioxide gas. How much CO_2 is made?

- 10.0 g = 5.60 g + mass of CO_2, so mass of CO_2 = **4.40 g**.
- In an open container the mass seems to drop, because the CO_2 escapes into the air. In a sealed container the mass stays exactly 10.0 g.

Parts of a chemical equation

$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

- **Reactants** on the left, **products** on the right; → means "yields", **+** separates substances.
- **Coefficients** (the big numbers in front, like the 2s) count units of each substance. A coefficient of 1 is not written.
- **Subscripts** (the small numbers, like the ₂ in H_2O) are part of the formula. **Changing a subscript makes a different substance**: H_2O_2 is hydrogen peroxide, not water.
- **State symbols:** (s) solid, (l) liquid, (g) gas, (aq) aqueous (dissolved in water).
- Conditions go over the arrow: Δ means heat is supplied; a catalyst's formula can be written there too.

The seven diatomic elements

Seven elements exist as two-atom molecules whenever they are on their own (not in a compound): **H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2**. Memory aid: **HOFBrINCl** (say “Hof-brin-kle”): H, O, F, Br, I, N, Cl. On the periodic table they make a “7” shape starting at element 7 (nitrogen), plus hydrogen.

So “oxygen gas” is always O_2, and “chlorine” as a reactant is always Cl_2.

From words to a balanced equation

1. **Word equation:** hydrogen + oxygen → water
2. **Skeleton equation** (correct formulas, not balanced yet): $H_2 + O_2 \rightarrow H_2O$
3. **Balanced equation:** $2H_2 + O_2 \rightarrow 2H_2O$

The balancing method

1. Write correct formulas. **Never change a subscript** after this step.
2. Make an **atom-count table** for both sides.
3. Balance **metals** first, then **nonmetals other than H and O**.
4. If a **polyatomic ion** (SO_4^{2-}, NO_3^-, PO_4^{3-} …) appears unchanged on both sides, balance it **as one unit**.
5. Balance **hydrogen**, then **oxygen** last (O often appears in several substances, so it is easiest at the end).
6. Re-count every atom. Make sure the coefficients are the **lowest whole numbers** (divide through if they share a factor).

Worked Example 1: $H_2 + O_2 \rightarrow H_2O$

Atom	Left	Right
H	2	2
O	2	1 ✗

- O is unbalanced. Put 2 in front of H_2O: $H_2 + O_2 \rightarrow \mathbf{2}H_2O$. Now O is 2 = 2, but H is 2 on the left and 4 on the right.
- Put 2 in front of H_2: $\mathbf{2}H_2 + O_2 \rightarrow 2H_2O$. H: 4 = 4, O: 2 = 2 ✓

Answer: $2H_2 + O_2 \rightarrow 2H_2O$

Worked Example 2: $Al + O_2 \rightarrow Al_2O_3$

- O: 2 on the left, 3 on the right. The lowest common multiple of 2 and 3 is **6**: put 3 in front of O_2 and 2 in front of Al_2O_3 → $Al + \mathbf{3}O_2 \rightarrow \mathbf{2}Al_2O_3$.
- Al: now 4 on the right, so put 4 in front of Al.

Atom	Left	Right
Al	4	2 × 2 = 4 ✓
O	3 × 2 = 6	2 × 3 = 6 ✓

Answer: $4Al + 3O_2 \rightarrow 2Al_2O_3$

Worked Example 3: $Pb(NO_3)_2 + KI \rightarrow PbI_2 + KNO_3$ (keep NO_3 as one unit)

- Pb: 1 = 1 ✓. NO_3 is unchanged on both sides, so count it as a unit: 2 on the left, 1 on the right. Put 2 in front of KNO_3.
- Now K is 2 on the right, so put 2 in front of KI. I: 2 = 2 ✓

Unit	Left	Right
Pb	1	1 ✓
NO_3	2	2 ✓
K	2	2 ✓
I	2	2 ✓

Answer: $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$. This is the yellow-precipitate reaction from Part 1.

Worked Example 4: combustion of propane, $C_3H_8 + O_2 \rightarrow CO_2 + H_2O$

- **C first:** 3 on the left, so $3CO_2$.
- **H next:** 8 on the left, so $4H_2O$ ($4 \times 2 = 8$).
- **O last:** right side has $3 \times 2 + 4 \times 1 = 10$ O atoms, so we need $5O_2$.

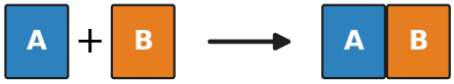
Atom	Left	Right
C	3	3 ✓
H	8	$4 \times 2 = 8$ ✓
O	$5 \times 2 = 10$	$3 \times 2 + 4 = 10$ ✓

Answer: $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

The fraction trick (ethane). $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$. C gives $2CO_2$, H gives $3H_2O$, so the right side has $2 \times 2 + 3 = 7$ O atoms. That needs 7/2 O_2, a fraction. Write $C_2H_6 + 7/2\ O_2 \rightarrow 2CO_2 + 3H_2O$, then **double everything**: **$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$**.

Part 3 — The Five Reaction Types (15 min)

Synthesis



Example: $2Na + Cl_2 \rightarrow 2NaCl$

Decomposition



Example: $2H_2O \rightarrow 2H_2 + O_2$

Single replacement



Example: $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ (A = Zn, B = Cu, C = SO_4)

Double replacement



Example: $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$ (partners swap)

Combustion

hydrocarbon + O_2 → $CO_2 + H_2O$ (gives off heat and light)

Example: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

1. Synthesis (combination): A + B → AB

Two or more substances combine into **one** product.

- $2Na + Cl_2 \rightarrow 2NaCl$ (metal + nonmetal gives an ionic compound)
- $CaO + H_2O \rightarrow Ca(OH)_2$ (metal oxide + water gives a base)

Predicting: metal + nonmetal → ionic compound. Use the ion charges to write its formula, then balance.

2. Decomposition: AB → A + B

One reactant breaks into two or more products, usually when energy (heat, electricity, light) is supplied.

- $2H_2O \rightarrow 2H_2 + O_2$ (electrolysis)
- $CaCO_3 \rightarrow CaO + CO_2$ (heating a metal carbonate gives a metal oxide + carbon dioxide)
- $2KClO_3 \rightarrow 2KCl + 3O_2$ (heating a chlorate gives a chloride + oxygen)

3. Single replacement (displacement): A + BC → AC + B

An **element** replaces another element in a compound. A metal replaces a metal (or hydrogen in an acid); a halogen replaces a halogen.

It only happens if the free element is more reactive than the one it replaces. The activity series ranks them:

Metals, most to least reactive	Halogens
Li > K > Ba > Ca > Na > Mg > Al > Zn > Fe > Ni > Sn > Pb > **(H)** > Cu > Ag > Au	F_2 > Cl_2 > Br_2 > I_2

- **Zn + $CuSO_4$ → $ZnSO_4$ + Cu.** Zn is above Cu, so it reacts: the blue solution fades and copper metal coats the zinc.
- **Cu + $ZnSO_4$ → no reaction.** Cu is below Zn.
- **Mg + 2HCl → $MgCl_2$ + H_2.** Metals above (H) replace hydrogen in acids and give hydrogen gas. Cu, Ag and Au, below H, do not react with HCl.
- **Cl_2 + 2NaBr → 2NaCl + Br_2.** Cl_2 is above Br_2.

Why the series works. Reacting means losing electrons (for metals). A metal higher in the series gives up its electrons more easily, so it ends up as the ion and pushes the less reactive metal out of solution as the element. This links back to Unit 2: down a group, ionization energy falls and the metals become more reactive (Na reacts faster than Li, K faster than Na).

4. Double replacement (double displacement): AB + CD → AD + CB

Two ionic compounds in solution **swap partners**. The positive ion of each pairs with the negative ion of the other.

It only happens if one product is a **precipitate**, a **gas**, or **water**. Otherwise all the ions just stay dissolved and nothing changes.

- **Precipitate:** $AgNO_3(aq)$ + NaCl(aq) → AgCl(s) + $NaNO_3(aq)$
- **Water (neutralization):** HCl(aq) + NaOH(aq) → NaCl(aq) + $H_2O(l)$
- **Gas:** Na_2CO_3 + 2HCl → 2NaCl + H_2O + CO_2 (the H_2CO_3 formed breaks down into H_2O and CO_2)

Short solubility rules (to decide whether a product is a precipitate):

Usually soluble (stays aqueous)	Exceptions
all Group 1 (Li^+, Na^+, K^+) and NH_4^+ compounds	none
all nitrates (NO_3^-) and acetates (CH_3COO^-)	none
chlorides, bromides, iodides	Ag^+, Pb^{2+}, Hg_2^{2+} compounds are insoluble
sulfates (SO_4^{2-})	$BaSO_4$, $PbSO_4$, $CaSO_4$ (slightly), $SrSO_4$
Usually insoluble (precipitate)	**Exceptions**
carbonates, phosphates, sulfides	Group 1 and NH_4^+ compounds are soluble
hydroxides (OH^-)	Group 1 soluble; $Ca(OH)_2$ and $Ba(OH)_2$ slightly soluble

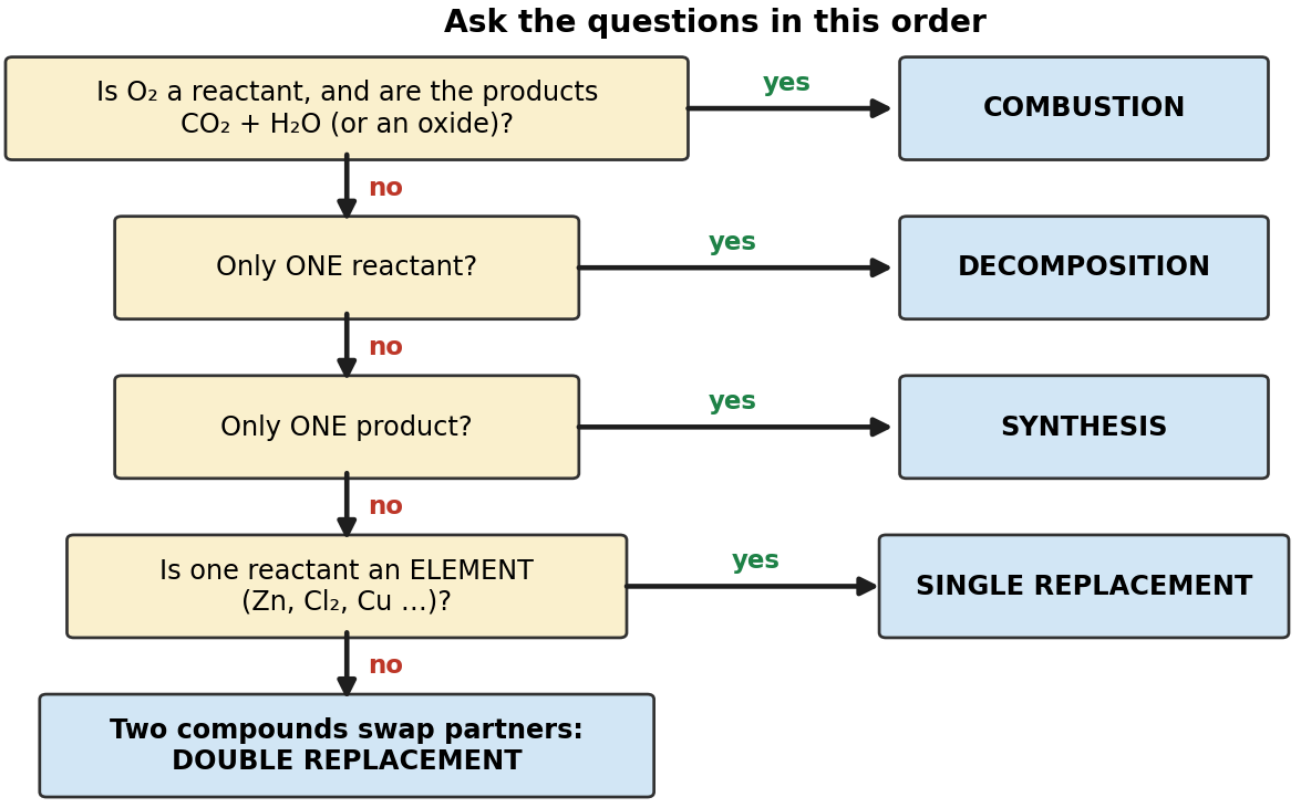
5. Combustion: fuel + O_2 → CO_2 + H_2O

A substance reacts rapidly with **oxygen**, releasing heat and light. For a **hydrocarbon** (C and H only) the products are always **CO_2 and H_2O** when there is plenty of oxygen.

- CH_4 + 2O_2 → CO_2 + 2H_2O (natural gas)
- With too little oxygen, **incomplete combustion** gives carbon monoxide (CO, a poisonous gas) and soot (C) instead of CO_2.

Overlap: 2Mg + O_2 → 2MgO is a combustion (Mg burns with a bright white flame) **and** a synthesis (one product). Your teacher may accept either, but name both if you aren't sure.

How to identify the type



Exit Ticket (3 min)

1. Balance: Fe + O_2 → Fe_2O_3
2. Classify, then predict the products and balance: Mg + $CuCl_2$ → ?
3. Does silver react with hydrochloric acid? Why or why not?

Practice Problems

Show an atom-count table for every balancing question. Do Q7, Q12 and Q13 in class if time allows; the rest are homework.

Tier 1 — Balance

1. N_2 + H_2 → NH_3
2. Na + Cl_2 → NaCl
3. $KClO_3$ → KCl + O_2
4. Fe + Cl_2 → $FeCl_3$
5. CH_4 + O_2 → CO_2 + H_2O

Tier 2 — Balance (harder)

6. $Al + HCl \rightarrow AlCl_3 + H_2$
7. $Ca(OH)_2 + H_3PO_4 \rightarrow Ca_3(PO_4)_2 + H_2O$
8. $Fe_2O_3 + CO \rightarrow Fe + CO_2$
9. $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$
10. $C_6H_{12}O_6 + O_2 \rightarrow CO_2 + H_2O$ (cellular respiration)

Tier 3 — Classify, predict, react or not

11. **Classify** each reaction: (a) $2Mg + O_2 \rightarrow 2MgO$ (b) $2H_2O_2 \rightarrow 2H_2O + O_2$ (c) $Zn + 2AgNO_3 \rightarrow Zn(NO_3)_2 + 2Ag$ (d) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$ (e) $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$
12. **Predict the products and balance:** (a) $Li + O_2$ (b) $Mg + HCl$ (c) $AgNO_3 + KBr$ (d) $C_4H_{10} + O_2$ (e) $CaCO_3$, heated (f) $Cl_2 + KI$
13. **Will it react?** Use the activity series. If yes, write the balanced equation. (a) $Cu + AgNO_3$ (b) $Ag + CuSO_4$ (c) $Fe + HCl$ (iron forms Fe^{2+}) (d) $I_2 + NaCl$ (e) $Zn + Pb(NO_3)_2$
14. **Word equations:** write the balanced equation with state symbols, and name the type. (a) Solid sodium reacts with water to form aqueous sodium hydroxide and hydrogen gas. (b) Aqueous lead(II) nitrate reacts with aqueous potassium iodide to form solid lead(II) iodide and aqueous potassium nitrate. (c) Aluminum metal is placed in aqueous copper(II) chloride, forming aqueous aluminum chloride and copper metal.

Tier 4 — Reasoning and challenge

15. **Conservation of mass:** 2.43 g of magnesium burns completely in oxygen and forms 4.03 g of magnesium oxide. (a) What mass of oxygen reacted? (b) A student mixes baking soda and vinegar in an open beaker on a balance and sees the mass go down. Does this break the law of conservation of mass? Explain.
16. **Chemical or physical?** Give your reason for each. (a) Ice melts at 0 °C. (b) An Alka-Seltzer tablet fizzes in water. (c) Water boils and bubbles. (d) Two clear solutions are mixed and a yellow solid appears. (e) A light bulb filament glows and gets hot.
17. **Challenge — balance:** (a) $C_8H_{18} + O_2 \rightarrow CO_2 + H_2O$ (octane in gasoline) (b) $NH_3 + O_2 \rightarrow NO + H_2O$

Answer Key — Full Step-by-Step Solutions

Exit ticket

1. O: LCM of 2 and 3 is 6, so $3O_2$ and $2Fe_2O_3$; then Fe: 4 on the right, so 4Fe. **$4Fe + 3O_2 \rightarrow 2Fe_2O_3$** (Fe 4 = 4, O 6 = 6 ✓).
2. **Single replacement.** Mg is above Cu in the activity series, so Mg replaces Cu: **$Mg + CuCl_2 \rightarrow MgCl_2 + Cu$** (already balanced: Mg 1, Cu 1, Cl 2 on each side).
3. **No.** Silver is below hydrogen in the activity series, so it can't replace hydrogen in HCl.

1. $N_2 + 3H_2 \rightarrow 2NH_3$

- N: 2 on the left, so $2NH_3$. H: now 6 on the right, so $3H_2$.

Atom	Left	Right
N	2	2 ✓
H	6	6 ✓

2. $2Na + Cl_2 \rightarrow 2NaCl$

- Cl: 2 on the left, so 2NaCl. Na: now 2 on the right, so 2Na. Na 2 = 2, Cl 2 = 2 ✓

3. $2KClO_3 \rightarrow 2KCl + 3O_2$

- O: 3 on the left, 2 on the right; LCM 6, so $2KClO_3$ and $3O_2$. Then K and Cl: 2 on the left, so 2KCl.

Atom	Left	Right
K	2	2 ✓
Cl	2	2 ✓
O	6	6 ✓

4. $2Fe + 3Cl_2 \rightarrow 2FeCl_3$

- Cl: 2 on the left, 3 on the right; LCM 6, so $3Cl_2$ and $2FeCl_3$. Then Fe: 2Fe. Fe 2 = 2, Cl 6 = 6 ✓

5. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

- C: 1 = 1. H: 4 on the left, so $2H_2O$. O: right side 2 + 2 = 4, so $2O_2$.

Atom	Left	Right
C	1	1 ✓
H	4	4 ✓
O	4	4 ✓

6. $2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$

- Cl: 1 on the left, 3 on the right, so 3HCl … but then H is 3 on the left and 2 on the right. LCM of 3 and 2 is 6: 6HCl and $3H_2$. Now Cl is 6 on the left, so $2AlCl_3$; then 2Al.

Atom	Left	Right
Al	2	2 ✓
H	6	6 ✓
Cl	6	6 ✓

7. $3Ca(OH)_2 + 2H_3PO_4 \rightarrow Ca_3(PO_4)_2 + 6H_2O$

- Ca: 3 on the right, so $3Ca(OH)_2$. PO_4 as a unit: 2 on the right, so $2H_3PO_4$.
- H: left 3 × 2 + 2 × 3 = 12, so $6H_2O$. O (outside PO_4): left 3 × 2 = 6 (from OH), right 6 (from H_2O) ✓

Unit	Left	Right
Ca	3	3 ✓
PO_4	2	2 ✓
H	6 + 6 = 12	12 ✓

Unit	Left	Right
O in OH / H_2O	6	6 ✓

(This is a neutralization: an acid plus a base gives a salt plus water.)

8. $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

- Fe: 2Fe. Each CO picks up one O from Fe_2O_3 to become CO_2, and Fe_2O_3 has 3 O to give, so 3CO and $3CO_2$.

Atom	Left	Right
Fe	2	2 ✓
C	3	3 ✓
O	3 + 3 = 6	3 × 2 = 6 ✓

9. $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$ (the fraction trick: 7/2 O_2, then double everything)

Atom	Left	Right
C	4	4 ✓
H	12	12 ✓
O	14	8 + 6 = 14 ✓

10. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

- C: $6CO_2$. H: 12, so $6H_2O$. O on the right: 12 + 6 = 18. Glucose already supplies 6, so O_2 must supply 12: $6O_2$.

Atom	Left	Right
C	6	6 ✓
H	12	12 ✓
O	6 + 12 = 18	12 + 6 = 18 ✓

11. Classify

- **(a)** Synthesis (one product); also the combustion of magnesium.
- **(b)** Decomposition (one reactant).
- **(c)** Single replacement (the element Zn replaces Ag).
- **(d)** Double replacement ($BaSO_4$ is the precipitate).
- **(e)** Combustion (hydrocarbon + $O_2 \rightarrow CO_2 + H_2O$).

12. Predict and balance

- **(a)** Synthesis: Li^+ and O^{2-} give Li_2O. **$4Li + O_2 \rightarrow 2Li_2O$**
- **(b)** Single replacement: Mg is above H. **$Mg + 2HCl \rightarrow MgCl_2 + H_2$**
- **(c)** Double replacement: AgBr is insoluble (Ag^+ halide). **$AgNO_3 + KBr \rightarrow AgBr(s) + KNO_3$**
- **(d)** Combustion: C gives $4CO_2$, H gives $5H_2O$, so O = 8 + 5 = 13, i.e. 13/2 O_2; double: **$2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$**
- **(e)** Decomposition of a carbonate: **$CaCO_3 \rightarrow CaO + CO_2$**
- **(f)** Single replacement: Cl_2 is above I_2. **$Cl_2 + 2KI \rightarrow 2KCl + I_2$**

13. Will it react?

- **(a) Yes**, Cu is above Ag: **$Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$**
- **(b) No**, Ag is below Cu.
- **(c) Yes**, Fe is above H: **$Fe + 2HCl \rightarrow FeCl_2 + H_2$**
- **(d) No**, I_2 is below Cl_2, so it can't replace chloride.
- **(e) Yes**, Zn is above Pb: **$Zn + Pb(NO_3)_2 \rightarrow Zn(NO_3)_2 + Pb$**

14. Word equations

- **(a)** $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$. **Single replacement** (Na replaces one H in water). Check: Na 2 = 2, H 4 = 2 + 2, O 2 = 2 ✓
- **(b)** $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$. **Double replacement** (Worked Example 3).
- **(c)** $2Al(s) + 3CuCl_2(aq) \rightarrow 2AlCl_3(aq) + 3Cu(s)$. **Single replacement.** Cl: LCM of 2 and 3 is 6, so $3CuCl_2$ and $2AlCl_3$; then 2Al and 3Cu. Al 2 = 2, Cu 3 = 3, Cl 6 = 6 ✓

15. Conservation of mass

- **(a)** mass of Mg + mass of O_2 = mass of MgO, so mass of O_2 = 4.03 − 2.43 = **1.60 g**.
- **(b) No.** The reaction makes CO_2 gas, which escapes from the open beaker, so the balance doesn't weigh it. In a sealed container the total mass stays the same.

16. Chemical or physical?

- **(a) Physical:** still H_2O, only the state changes.
- **(b) Chemical:** a new gas (CO_2) is produced by a reaction.
- **(c) Physical:** the bubbles are water vapor, the same substance.
- **(d) Chemical:** a precipitate, a new solid substance (for example PbI_2).
- **(e) Physical:** the filament is still tungsten; electrical energy becomes heat and light without a new substance.

17. Challenge

- **(a)** C: $8CO_2$. H: 18, so $9H_2O$. O on the right: 16 + 9 = 25, so 25/2 O_2. Double everything: **$2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$** (C 16, H 36, O 50 on each side).
- **(b)** H: 3 in NH_3 and 2 in H_2O; LCM 6, so $2NH_3$ and $3H_2O$. N: 2NO. O on the right: 2 + 3 = 5, so 5/2 O_2. Double everything: **$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$** (N 4, H 12, O 10 on each side).

The 8 mistakes that cost the most marks

1. **Changing a subscript to balance** (writing H_2O_2 to fix the oxygen). Only coefficients change.
2. **Forgetting diatomic elements:** writing O or Cl as a reactant instead of O_2 or Cl_2.
3. **Wrong formulas before balancing:** e.g. AlCl instead of $AlCl_3$. Get the charges right first.
4. **Splitting a polyatomic ion** that appears unchanged on both sides, which makes the counting much harder.
5. **Leaving coefficients that can be divided**, like $4H_2 + 2O_2 \rightarrow 4H_2O$. Reduce to $2H_2 + O_2 \rightarrow 2H_2O$.
6. **Writing a single replacement that can't happen.** Check the activity series first.

7. **Writing a double replacement with no driving force.** If no precipitate, gas or water forms, there is no reaction.
8. **Treating evidence as proof:** bubbles from boiling water are not a chemical change.

Notes for next session

- **C7 — Predicting products in depth:** more practice with the activity series and solubility rules, then **net ionic equations** and **spectator ions** (a Gifted-level topic in this unit). Example: $AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$ becomes $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$, because Na^+ and NO_3^- don't change.
- **C8 — Oxidation numbers and redox:** which atoms lose and gain electrons in single replacement and combustion, then a Unit 4 review and practice test.
- **Then Unit 5, stoichiometry:** the mole (Avogadro's number, molar mass, conversions), percent composition and empirical formulas, mole ratios and mass-to-mass problems, limiting reactant and percent yield. Today's coefficients will be read as **mole ratios**.